How-To Library

Informatica

Using the Blaze Engine to Run Profiles and Scorecards

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Abstract

Informatica Blaze engine is an Informatica proprietary engine for distributed processing on Hadoop. You can run profiles and scorecards on the Blaze engine. This article discusses how you can use the Blaze engine to run profiles and scorecards in Informatica Developer and Informatica Analyst.

Supported Versions

- Data Quality 10.1
- Big Data Management 10.1

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Blaze Engine Overview

You can use the Blaze engine to run profiles or scorecards on data sources with large volume of data, on a variety of data sources, and on Big data. The Blaze engine increases performance and scalability of the Hadoop cluster.

The Blaze engine is a part of Informatica Big Data Management. After you download and install Informatica Big Data Management, you can use the Blaze engine to run profiles and scorecards in Informatica Developer and Informatica Analyst.

The Blaze engine is built using a memory-based data exchange framework which runs natively on YARN without the dependence of MapReduce or Hive. The Blaze distributed processing engine has the ability to scale and perform high-speed data processing of large complex batch workloads using a natively-embedded Informatica data transformation engine on Hadoop. The profiles and scorecards are processed faster and the results are retrieved quickly from the cluster.

To run a profile or scorecard on the Blaze engine, the Data Integration Service submits jobs to the Blaze engine executor. The Blaze engine executor is a software component that enables communication between the Data Integration Service and the Blaze engine components on the Hadoop cluster. You can create the Hadoop connections using the Developer tool, Administrator tool, and infacmd.

You can use the Blaze engine to run column profiles and enterprise discovery profiles in the Developer tool. You can use the Blaze engine to run column profiles, enterprise discovery profiles, and scorecards in the Analyst tool.

Profiles and Scorecards on the Blaze Engine

You can run profiles or scorecards in the Hadoop environment on the Blaze engine after you install Informatica Big Data Management.

When you run the profiles or scorecards in the Hadoop run-time environment on the Blaze engine, the following process occurs:

1. The Analyst tool or Developer tool submits the job to the Profiling Service Module.
2. The Profiling Service Module breaks down the job into a set of mappings.

3. The Data Integration Service pushes the mapping execution to the Hadoop cluster through a Hadoop connection.
4. The profile results or scorecard results are saved in the profiling warehouse.
5. You can view the profile results in the Analyst tool or Developer tool and view the scorecard results in the Analyst tool.

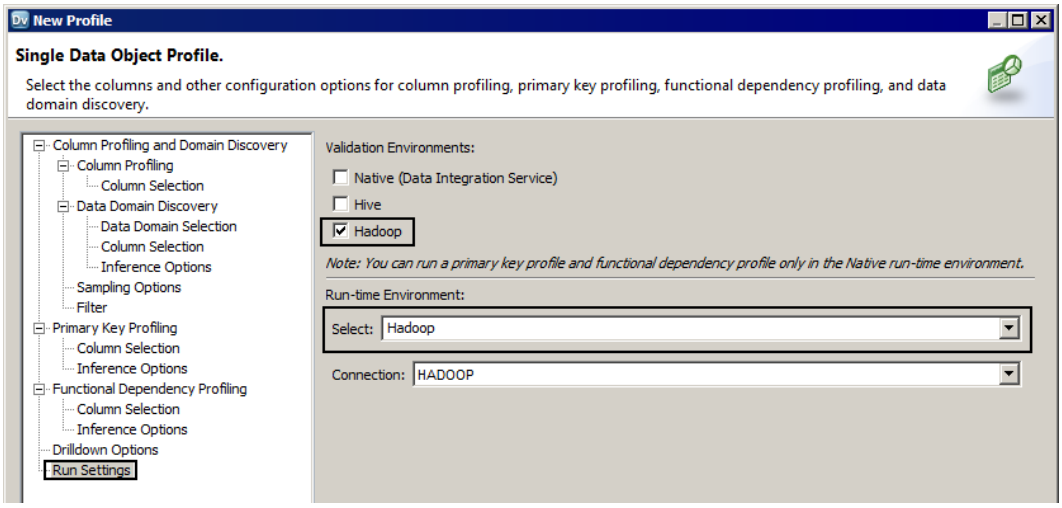
Creating a Column Profile in Informatica Developer

You can choose to run column profiles and enterprise discovery profiles in Informatica Developer. In the Developer tool, column profiles include single data object profiles and multiple data object profiles.

To create a single data object profile, perform the following steps:

1. In the **Object Explorer** view, select the data object you want to profile.
2. Click **File** > **New** > **Profile** to open the profile wizard.
3. Select **Profile** and click **Next** .
4. Enter a name for the profile and verify the project location. Optionally, enter a text description of the profile. Click **Next**.
5. Review and edit the column selection, filter and sampling options, inference options, drill-down options, and data domain selection as per your requirements.
6. In the **Run Settings** section:
   - Choose **Hadoop** as the validation environment.
   - Choose **Hadoop** to use the Blaze engine to run the profile and select a Hadoop connection.

   The following image shows the run-time environment options that you can choose to run a column profile in the Developer tool:



7. Click **Finish**.

   The Blaze engine runs the profile and the profile results appear in the Developer tool.

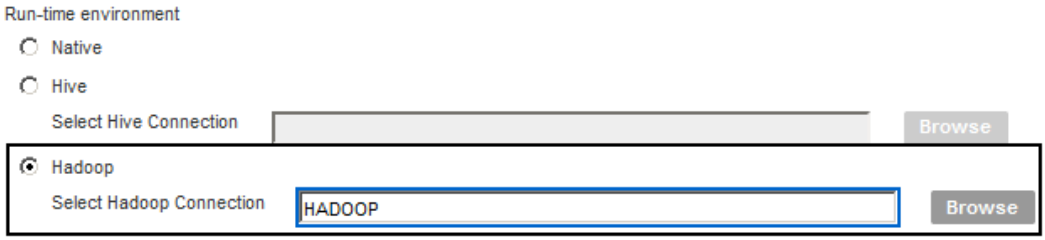
Creating and Running a Column Profile in Informatica Analyst

You can create and run column profiles and enterprise discovery profiles on the Blaze engine in the Analyst tool.

To create a column profile in the Analyst tool, perform the following steps:

1. In the **Discovery** workspace, select **New** > **Profile** from the header area.
2. The **Single source** option is selected by default. Click **Next**.
3. Enter information about the profile in **Specify General Properties** screen and **Select Source** screen. Click **Next**
4. In the **Specify Settings** screen, choose the options, as required.
5. Choose **Hadoop** as the run-time environment. Click **Browse** to select a Hadoop connection in the **Select a Hadoop Connection** dialog box.

   The following image shows the run-time environment options that you can choose to run a column profile in the Analyst tool:



6. In the **Specify Rules and Filters** screen, create, edit, or delete a rule or filer, as required.
7. Click **Save and Run** to create and run the profile.

   The Blaze engine runs the profile and the profile results appears in the summary view.

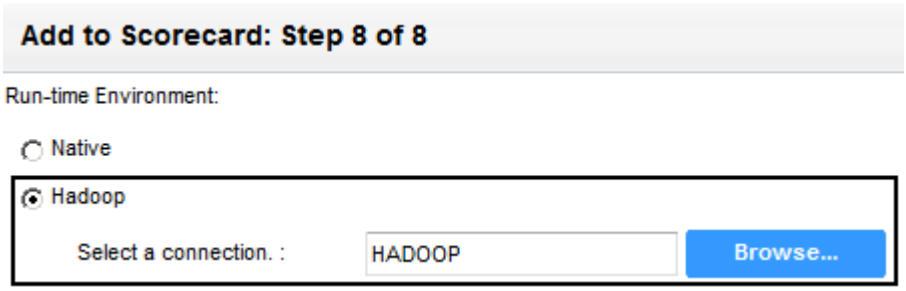
Creating and Running a Scorecard in Informatica Analyst

You can create and run scorecards in Informatica Analyst. You can also create scorecards in Informatica Developer and run the scorecards in the Analyst tool. You can use the Blaze engine to run scorecards.

1. In the **Library** workspace, select the project or folder that contains the profile.
2. Click the profile to open the profile.

   The profile results appear in the summary view in the **Discovery** workspace.
3. The **Add to Scorecard** wizard appears
4. In the **Add to Scorecard** screen, choose to create a new scorecard. Click **Next** .
5. In the **Step 2 of 8** screen, enter a name for the scorecard and a description for the scorecard. Select the project and folder where you want to save the scorecard. Click **Next**
6. Enter information from **Step 3 of 8** screen through **Step 7 of 8** screen, as required. Click **Next** .
7. In the **Step 8 of 8** screen, choose **Hadoop** as the run-time environment. Click **Choose** to select a Hadoop connection in the **Select a Hadoop Connection** dialog box.

   The following image shows the run-time environment options that you can choose to run the scorecard:



8. Click **Save & Run** to save and run the scorecard.

   The Blaze engine runs the scorecard in the Hadoop cluster and the scorecard results appears in the **Scorecard** workspace.

Author

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